

# 1 Model

For our example, we adapt the model from Wolkovich & Donahue 2020, which is itself an adaptation of a common model of community dynamics, often conceptualized for annual plants with a seedbank. A more detailed description of the model can be found in Wolkovich & Donahue 2020.

We model a pair of competing species in which the seedbank  $N$  of species  $i$  at time  $T + 1$  (where  $T$  is the beginning of an annual growing season) is determined by the survival ( $s$ ) of ungerminated seeds from the previous year ( $1 - g_i(T)$ ) plus new biomass ( $B_i$ ) produced during the growing season (from  $T$  to  $T + \delta$ ), which is converted to seeds at rate  $\phi$ :

$$N_i(T + 1) = sN_i(T)(1 - g_i(T)) + \phi B_i(T + \delta) \quad (1)$$

Within season dynamics of biomass are modeled as  $R^*$  competition that begins when resources become available at the beginning of the growing season,  $R(t = 0) = R_0(T)$ , where  $t$  indicates within-season time and annual initial resource availability. For each species, biomass increases through germination  $G_i(t)$  and resource uptake ( $f(R)$ ), which is converted into biomass at a species-specific rate ( $c_i$ ), less maintenance costs ( $m$ ) (Figure S1b,c). Resource uptake per unit biomass is defined by the change in uptake per unit resource ( $a$ ) and the maximum uptake ( $1/u$ ).

$$\frac{dB_i}{dt} = (c_i f(R) - m)B_i + G_i(t) \quad (2)$$

$$f(R) = \frac{aR}{1 + auR} \quad (3)$$

The resource ( $R$ ) itself declines across a growing season (Figure S1a) due to uptake by the standing biomass of both species and abiotic loss ( $\epsilon$ ):

$$\frac{dR}{dt} = - \sum_{i=1}^2 f(R)B_i - \epsilon R \quad (4)$$

Our model departs from similar community models by explicit modeling of within-season germination phenology: germination, ( $G_i(t)$ ), is modeled as a time-dependent forcing function that adds daily biomass (germinants) according to a Hill equation where the timing of 50%-germination ( $\tau_{g50,i}$ ) differs by species.

$$G_i(t) = N_0 s_i b_i g_i(t) \quad (5)$$

$$g_i(t) = g_{max} \left( \frac{t^h}{\tau_{g50,i}^h + t^h} \right) \quad (6)$$

where  $N_0$  is the number of overwintering seeds,  $s_i$  is overwinter seed survival,  $b_i$  converts germinating seeds into biomass, and  $g_i(t)$  is the percent germination at time ( $t$ ) (Figure S1e,f). The shape of the Hill Equation defining  $g_i(t)$  has the same maximum germination ( $g_{max}$ ) and Hill coefficient ( $h$ ) for each species, and  $\tau_{g50,i}$  is the species-specific delay due to overwinter chilling (Figure S2a,b). The number of days of germination delay for species  $i$ ,  $\tau_{delay,i}$ , declines with chilling, such that

$$\tau_{delay,i} = \tau_{max} e^{-\tau_{\xi,i} \xi(T)} \quad (7)$$

$$\tau_{g50,i} \sim \text{Poisson}(\tau_{delay,i}) \quad (8)$$

where  $\xi(T)$  is the weeks of overwinter chilling in year  $T$ ,  $\tau_{\xi,i}$  is the species-specific sensitivity to chilling, and  $\tau_{max}$  is the expected delay when overwinter chilling is zero. That is, germination depends on the interaction of a species trait and a varying environment, resulting in year to year variation in the speed of germination (Figure S2e) and the timing of peak germination (Figure S2f).

#### *Simulations:*

To examine species coexistence when the timing of germination depends on annually varying seed stratification, we ran 10000 simulations of two-species communities. Communities were assembled from randomly generated species, where each species varied in resource conversion efficiency ( $c_i$ ) and in sensitivity to chilling ( $\tau_{\xi,i}$ ) (see Table S1 for parameter values). Variation in resource conversion efficiency yields different  $R^*$  values, a metric of resource competition where a lower  $R^*$  means a species can draw the resource down to a lower level and is thus considered the superior competitor. Variation in sensitivity to chilling resulted in variation in the distribution of germination within a growing season (Figure S1d). All other parameters were identical between species (Table S1). Within-year competitive dynamics were solved using an ode solver (ode in the R package `deSolve`) with initial condition  $R = R_0(T)$  and  $B_i(t = 0) = 0$ , and germination was as a forcing function with germinants for each species added daily in proportion to a Hill Equation (Eqn 6, Figure S1d). The end-of-season biomass of each species was converted to seeds, and the populations were censused. At each census, a minimum cutoff was applied to define extinction from the model. Note that ‘coexistence’ in this model is defined by joint persistence through time and not by low density growth rate.

In all simulations, the environment varied from year to year in both resource availability ( $R_0(T)$ ) and overwinter chilling ( $\xi(T)$ ), such that

$$R_0(T) \sim \log N(\mu_R, \sigma_R) \quad (9)$$

$$(\xi(T) \sim \log N(\mu_\xi, \sigma_\xi)) \quad (10)$$

Variation in  $R_0(T)$  and  $\xi(T)$  were independent. We compared competitive coexistence in simulations from two different environmental regimes: a low-chill regime and a high-chill regime (Figure 2a, Table S1). Each species was initialized with a seedbank ( $N_i(0) = 100$ ), and simulations were run for 300 years. For species that coexisted in the high-chill regime, we repeated the simulation of two species with the same resource conversion efficiency ( $c_i$ ) and sensitivity to chilling ( $\tau_{\xi,i}$ ) in a low-chill regime.

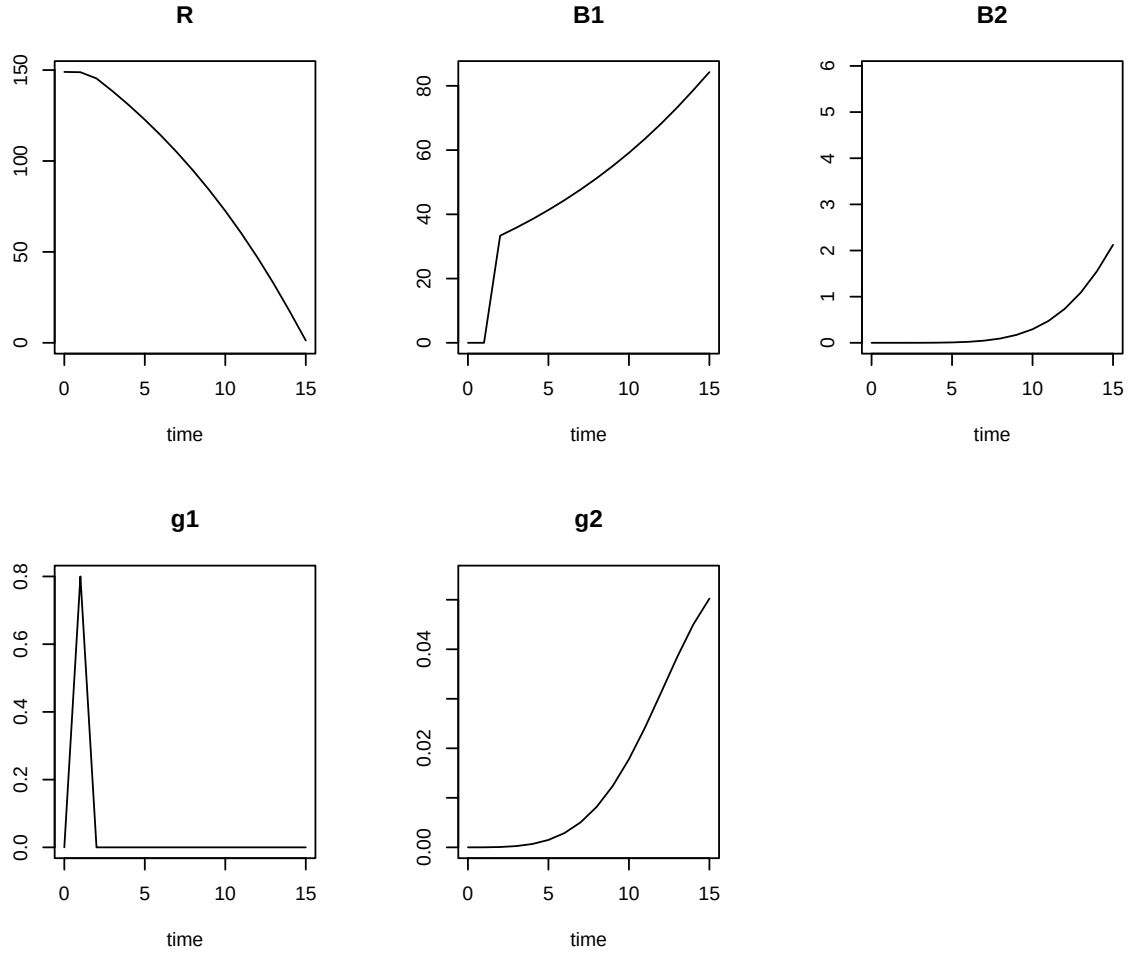


Figure S1: Within-season dynamics of (a) resources ( $R(t)$ , Eqn 3), (b,c) biomass of two competing species ( $B_i(t)$ , Eqn 2), and (d,e) the distribution of germination over the growing season for each species ( $g_i(t)$ , Eqn 6)

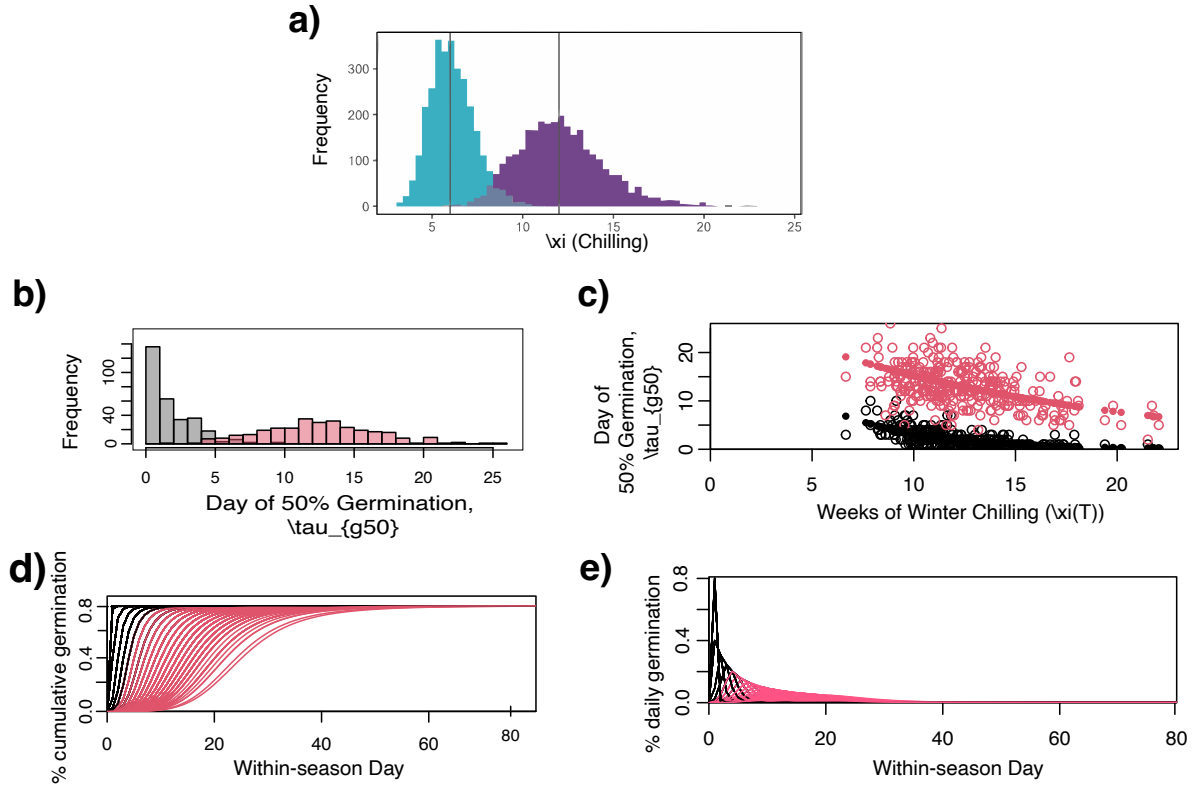


Figure S2: Environmental variation and species-specific responses. (a) Comparison of the distribution in winter chilling in low-chill and high-chill regimes; (b) Species-specific distribution of annual variation in the timing of 50%-germination; (c) Species-specific response to chilling in the timing of 50%-germination; (d) Species-specific patterns in daily germination over the growing season in year T; (e) Species-specific patterns in cumulative germination over the growing season in year T. Panels b and c are plotted for a high-chill regime

Table S1: Parameter values, definitions, and units.

| Parameter              | Value(s)                                                                                                                    | Definition                                                                                            | Unit                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| $R_0(T)$               | $\sim \log N(\mu_R, \sigma_R), \mu_R = \log(2), \sigma_R = 0.2$                                                             | amount of resource at the beginning of the growing season, varies annually                            | resource                                                                                       |
| $\xi(T)$               | $\sim \log N(\mu_\xi, \sigma_\xi), \mu_{\xi,lo} = \log(6), \sigma_\xi = 0.2$ or $\mu_{\xi,hi} = \log(12), \sigma_\xi = 0.2$ | annually varying winter chilling                                                                      | weeks of chilling                                                                              |
| $N_i(T)$               | $N_i(T = 0) = 100; \min(N_i(T)) = 10^{-4}$                                                                                  | census of seedbank of species $i$                                                                     | seeds per unit area                                                                            |
| $s$                    | 0.8                                                                                                                         | overwinter survival of species $i$                                                                    | unitless                                                                                       |
| $\phi$                 | 0.05                                                                                                                        | conversion of end-of-season biomass to seeds                                                          | $\text{biomass}^{-1}$ , but conceptually $\frac{\text{seeds}}{(\text{biomass})(\text{seeds})}$ |
| $B_i(t)$               | $B_i(t = 0) = G_i(t = 0)$ ; see Eqns 2,5                                                                                    | within-season biomass of species $i$ , initiated by germination                                       | biomass                                                                                        |
| $c_i$                  | $\sim \text{Unif}(0.025, 0.75)$                                                                                             | conversion efficiency of $R$ to biomass of species $i$                                                | $\frac{\text{biomass}}{\text{resource}}$                                                       |
| $m$                    | 0.005                                                                                                                       | maintenance costs during growth season for species $i$                                                | $\text{days}^{-1}$                                                                             |
| $a$                    | 0.8                                                                                                                         | uptake increase as $R$ increases for species $i$                                                      | $\text{days}^{-1}$                                                                             |
| $u$                    | 5                                                                                                                           | inverse of maximum uptake for species $i$                                                             | $\frac{(\text{days})(\text{biomass})}{\text{resource}}$                                        |
| $\theta$               | 1                                                                                                                           | shape of uptake for species $i$                                                                       | unitless                                                                                       |
| $\epsilon$             | 1                                                                                                                           | abiotic loss of $R$                                                                                   | $\text{days}^{-1}$                                                                             |
| $g_{max}$              | 0.5                                                                                                                         | maximum germination rate of species                                                                   | unitless                                                                                       |
| $h$                    | 100                                                                                                                         | controls the the rate at which germination declines as $\tau_p$ deviates from optimum for species $i$ | $\text{days}^{-2}$                                                                             |
| $g_i$                  | see Eqn 6                                                                                                                   | germination rate                                                                                      | unitless                                                                                       |
| $\tau_g$               | $\sim \beta(?, ?)$                                                                                                          | days of delay (given on weeks of chilling)                                                            | days                                                                                           |
| $\tau_{g50}$           | $\sim \text{Pois}(\tau_g)$                                                                                                  | ??                                                                                                    | days                                                                                           |
| $\xi_\tau$             | $\sim ?, ?$                                                                                                                 | rate of delay (given weeks of chilling) $i$                                                           | days/week                                                                                      |
| $b_i$                  | 1                                                                                                                           | biomass of a seedling                                                                                 | $\frac{\text{biomass}}{\text{seeds}}$                                                          |
| $f(R)$                 | see Eqn 3                                                                                                                   | resource uptake rate for species $i$                                                                  | $\frac{\text{resource}}{(\text{days})(\text{biomass})}$                                        |
| $t$                    | 1                                                                                                                           | annual timestep                                                                                       | years                                                                                          |
| $0 \rightarrow \delta$ | determined by rate of resource depletion                                                                                    | time during the growing season                                                                        | days                                                                                           |